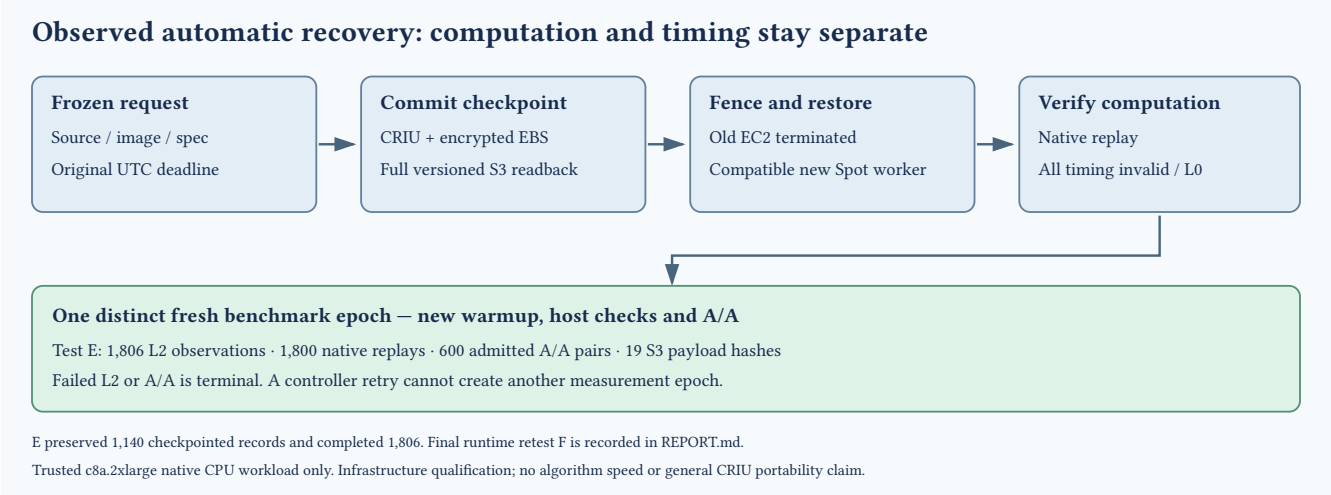
Automatic CRIU / EBS recovery on EKS

Iso Lab · 21 October 2026 · native harness qualification

The AWS-only native harness survived a stopped controller and the deliberate loss of its own Spot worker. Test E preserved a 1,140-record checkpoint prefix and completed 1,806 records after restoration. Its separate fresh epoch passed **1,806 L2 observations, 1,800 native replays, 600 A/A pairs and 19 S3 payload hashes**. A duplicate terminal submission created no second epoch.



Scope and frozen workload

This qualifies infrastructure and correctness, not an algorithm speedup or arbitrary workload recovery. The public fixture uses prime_search bits 32 / seed 59297; two targets / seed 20261002; rho.negation, rho.plain and matching A/A; 300 measured rounds, one warmup. Test C used five rounds and completed before its first checkpoint. All changes were frozen before launch. Every resumed computation timing sample is invalid/L0; only the distinct fresh epoch earns L2.

The supported recovery class is c8a.2xlarge, AMD family 26/model 2/stepping 1, AL2023 kernel 6.18.54-121.166.amzn2023.x86_64, containerd 2.3.5+unknown and CRIU 4.2.1. Source, binary, runtime and request hashes bind every attempt. The original absolute deadline is never extended. Capacity stayed within the existing 96-vCPU Standard Spot quota and 64-vCPU lab cap.

Test E receipt	Observed value
Original → replacement worker	`i-0ad9c1eaaaf9033a7` → `i-0ae32d58bdccafdf4`
Encrypted snapshot	`snap-0af1b03e30d462fce`
Checkpoint readback	45,844,480 bytes; full versioned S3 SHA-256 verification
Original byte prefix	5,096,143 bytes / 1,140 records, preserved in final 1,806
Restored process tree	Root PID 531 and child PID 5133; stopped credential checks before SIGCONT
Fresh A/A interval	95%: 0.999546254387097–1.0006296636908705; 600 admitted pairs
Fresh Pod UID	`dffc2fc4-fb8f-48f7-92a6-e2d79d0e8178`

Failures, controls and remaining limits

A failed in bootstrap before computation; B was cancelled for a deployment provenance mismatch before any payload start; D was rejected by CodeBuild's single-controller limit. C passed the zero-restore path (36 L2 / 30 replays / 10 A/A / 19 hashes). These attempts remain recorded; none is a retried failed L2/A/A measurement. E's computation replay verifier ran under the supervisor; F passed the final runtime with replay observed at UID/GID 1000, empty groups, zero capabilities and no_new_privs. Original and restored payloads already ran with those restrictions. F preserved 1,147 records / 5,128,482 bytes and completed 1,806 computations across us-east-2a → us-east-2c. Its fresh epoch passed 1,806 L2 / 1,800 native replay / 600 A/A / 19 S3 hashes. A/A 95% interval: 0.9998302217833182–1.0003545982056818. The replacement exposed eight physical cores and eight logical CPUs, with no SMT. Its fresh Pod UID is `8a5f3da5-38b1-42d0-8ed5-1a51ef71720d`.

The supervisor remains privileged and is for trusted workloads. Ordinary benchmark Pods remain Restricted; containerd CRI restore-via-create is disabled. Sixteen native tests cover identity/deadline/fencing, incomplete generations, corrupt/truncated archive readback, credential checks and failed-epoch refusal. Live corrupt-snapshot multi-generation fallback, arbitrary commands, GPU/ARM and cross-family recovery are not qualified. The full main-only commit-building GitHub Action remains pending review/CI/merge; this AWS path does not prove it.

Source-linked details, final-runtime F results and all exact identities are in `REPORT.md` and `runs.tar.xz` (verify `SHA256SUMS` before extraction). The archive contains public records and receipts; raw process RAM stays in private encrypted S3. Canonical performance graphs and leaderboards remain unchanged because no speed claim is made.

The lab host MCP/client upgrade was also checked through a fresh initialized connection. Re-submitting completed F through MCP succeeded with the same request, attempt and benchmark UID. Disposable PVCs and the temporary admin grant were removed; encrypted checkpoints remain private.